

3. Trouble shooting

Step 1

Kernel – initramfs file remove

```
[root@client grub2]# cd ../
[root@client boot]# ls
config-5.14.0-697.el9.x86_64
config-5.14.0-710.el9.x86_64
efi
grub2
initramfs-0-rescue-e62ba35a032f4beb951d75b1feelf762.img
initramfs-5.14.0-697.el9.x86_64.img
initramfs-5.14.0-697.el9.x86_64kdump.img
initramfs-5.14.0-710.el9.x86_64.img
initramfs-5.14.0-710.el9.x86_64kdump.img
[root@client boot]# ll
total 523920
-rw-r--r--. 1 root root      240333 Apr 22 14:44 config-5.14.0-697.el9.x86_64
-rw-r--r--. 1 root root      240450 May 27 14:43 config-5.14.0-710.el9.x86_64
drwxr-xr-x. 3 root root         17 Jun 10 15:04 efi
drwx----- 5 root root         97 Jun 26 19:49 grub2
-rw----- 1 root root 268647766 Jun 10 15:13 initramfs-0-rescue-e62ba35a032f4beb951d75b1feelf762.img
-rw----- 1 root root 62981999 Jun 10 15:17 initramfs-5.14.0-697.el9.x86_64.img
-rw----- 1 root root 39354368 Jun 21 17:03 initramfs-5.14.0-697.el9.x86_64kdump.img
-rw----- 1 root root 60868039 Jun 22 11:18 initramfs-5.14.0-710.el9.x86_64.img
-rw----- 1 root root 39386112 Jun 22 17:55 initramfs-5.14.0-710.el9.x86_64kdump.img
drwxr-xr-x. 3 root root         21 Jun 10 15:08 loader
lrwxrwxrwx. 1 root root         45 Jun 10 15:10 symvers-5.14.0-697.el9.x86_64.gz -> /lib/modules/5.14.0-697.el9.x86_64/symvers.gz
lrwxrwxrwx. 1 root root         45 Jun 22 11:17 symvers-5.14.0-710.el9.x86_64.gz -> /lib/modules/5.14.0-710.el9.x86_64/symvers.gz
-rw----- 1 root root    9561583 Apr 22 14:44 System.map-5.14.0-697.el9.x86_64
-rw----- 1 root root    9566527 May 27 14:43 System.map-5.14.0-710.el9.x86_64
-rwxr-xr-x. 1 root root 15206440 Jun 10 15:11 vmlinuz-0-rescue-e62ba35a032f4beb951d75b1feelf762
-rwxr-xr-x. 1 root root 15206440 Apr 22 14:44 vmlinuz-5.14.0-697.el9.x86_64
-rwxr-xr-x. 1 root root 15214632 May 27 14:43 vmlinuz-5.14.0-710.el9.x86_64
```

```
[root@client boot]# rm -rf initramfs-5.14.0-697.el9.x86_64.img initramfs-5.14.0-697.el9.x86_64kdump.img
[root@client boot]# ll
total 423980
-rw-r--r--. 1 root root      240333 Apr 22 14:44 config-5.14.0-697.el9.x86_64
-rw-r--r--. 1 root root      240450 May 27 14:43 config-5.14.0-710.el9.x86_64
drwxr-xr-x. 3 root root         17 Jun 10 15:04 efi
drwx----- 5 root root         97 Jun 26 19:49 grub2
-rw----- 1 root root 268647766 Jun 10 15:13 initramfs-0-rescue-e62ba35a032f4beb951d75b1feelf762.img
-rw----- 1 root root    60868039 Jun 22 11:18 initramfs-5.14.0-710.el9.x86_64.img
-rw----- 1 root root 39386112 Jun 22 17:55 initramfs-5.14.0-710.el9.x86_64kdump.img
drwxr-xr-x. 3 root root         21 Jun 10 15:08 loader
lrwxrwxrwx. 1 root root         45 Jun 10 15:10 symvers-5.14.0-697.el9.x86_64.gz -> /lib/modules/5.14.0-697.el9.x86_64/s
lrwxrwxrwx. 1 root root         45 Jun 22 11:17 symvers-5.14.0-710.el9.x86_64.gz -> /lib/modules/5.14.0-710.el9.x86_64/s
-rw----- 1 root root    9561583 Apr 22 14:44 System.map-5.14.0-697.el9.x86_64
-rw----- 1 root root    9566527 May 27 14:43 System.map-5.14.0-710.el9.x86_64
-rwxr-xr-x. 1 root root 15206440 Jun 10 15:11 vmlinuz-0-rescue-e62ba35a032f4beb951d75b1feelf762
-rwxr-xr-x. 1 root root 15206440 Apr 22 14:44 vmlinuz-5.14.0-697.el9.x86_64
-rwxr-xr-x. 1 root root 15214632 May 27 14:43 vmlinuz-5.14.0-710.el9.x86_64
```

Step 2

After removing initramfs file I would get a error like this

```
error: .././grub-core/fs/fshelp.c:257:file
'/initramfs-5.14.0-710.el9.x86_64.img' not
found.
```

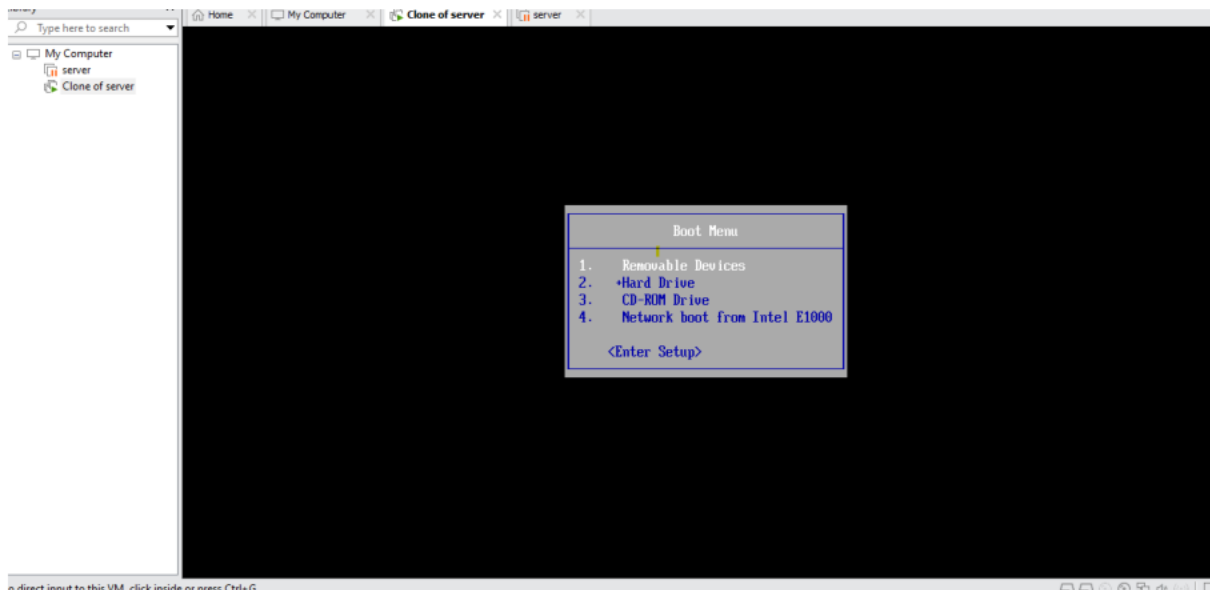
```
Press any key to continue...
```

Step 3

Reboot or shutdown then power on

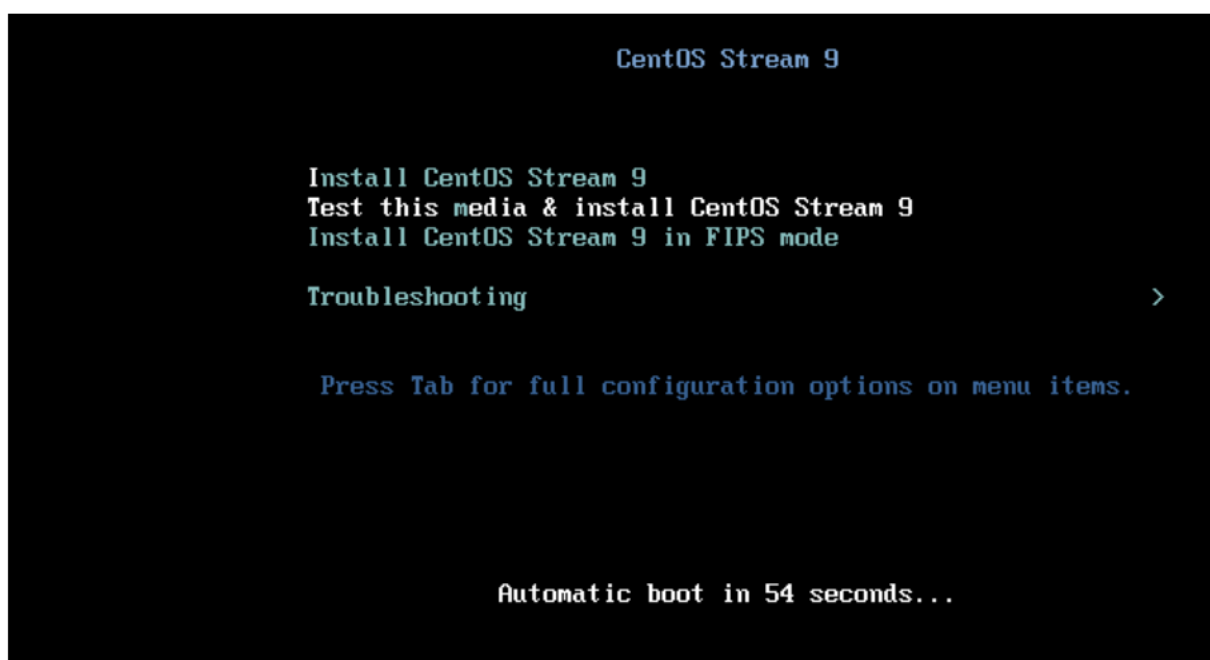
Now power on the machine with restart guest and press esc while screen splash

Click the cd rom drive

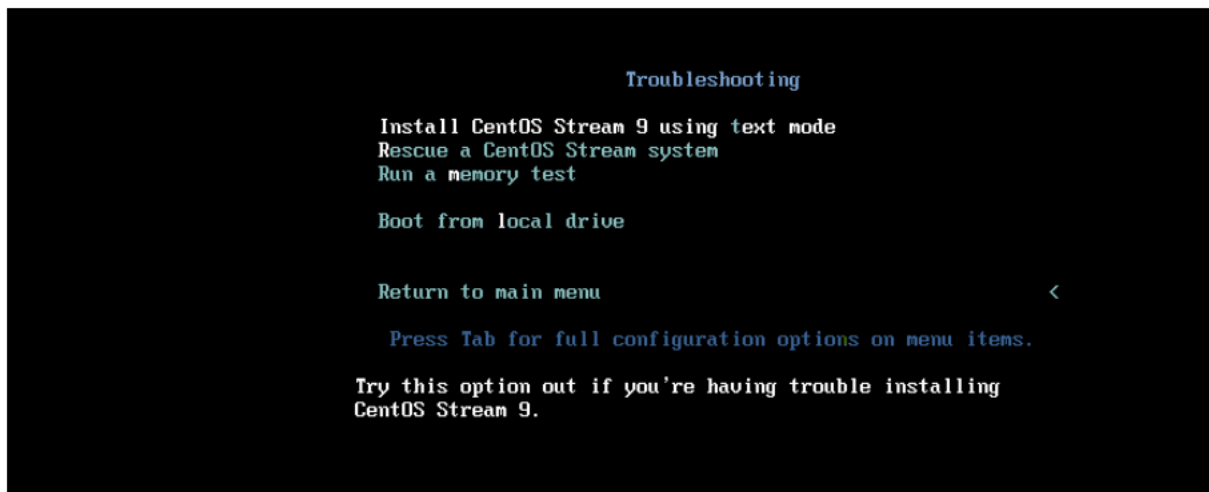


Step 4

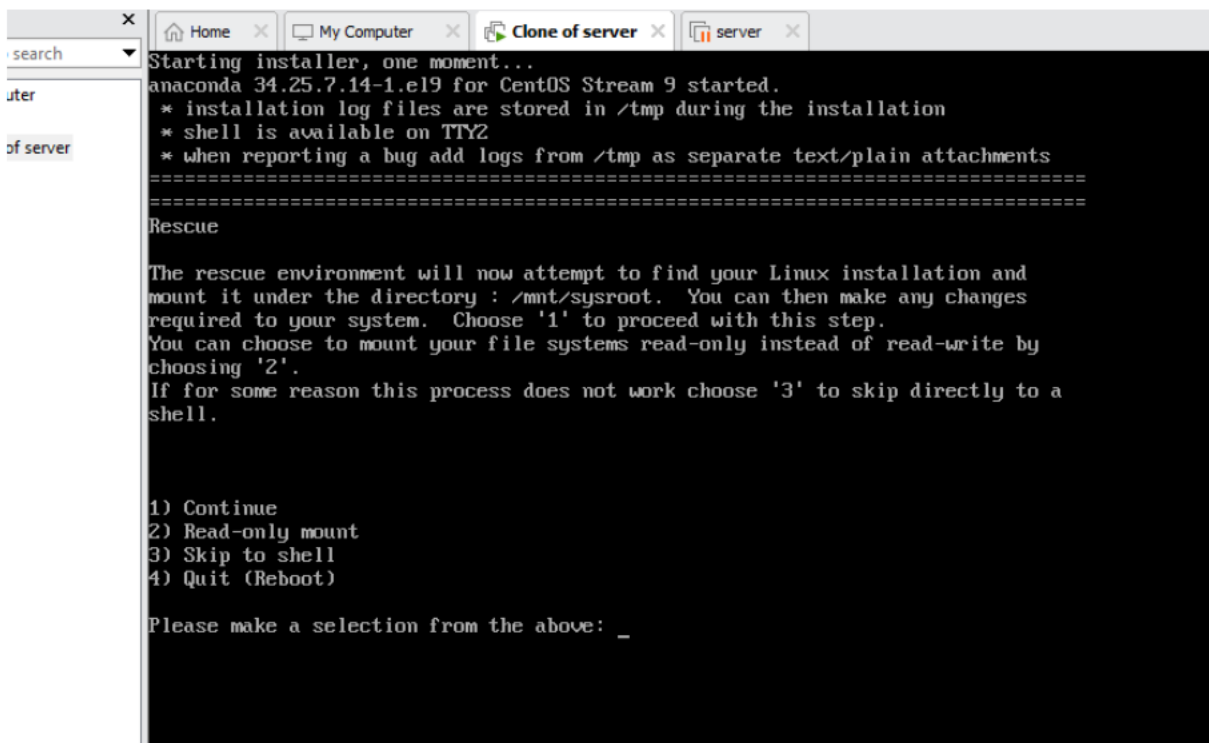
Click or select the troubleshooting to the grub error



Then select the rescue a centos stream system



It will take us to prombt where start 1 to continue



Then press enter to get shell

```
4) Quit (Reboot)

Please make a selection from the above: 1
=====
Rescue Shell

Your system has been mounted under /mnt/sysroot.

If you would like to make the root of your system the root of the active system,
run the command:

    chroot /mnt/sysroot

Warning: The rescue shell will trigger SELinux autorelabel on the subsequent
boot. Add "enforcing=0" on the kernel command line for autorelabel to work
properly.
When finished, please exit from the shell and your system will reboot.

Please press ENTER to get a shell:
```

Type the command

#Chroot /mnt/sysroot

#Dracut --regenerate-all --force

#exit

#exit

```
2) Read-only mount
3) Skip to shell
4) Quit (Reboot)

Please make a selection from the above: 1
=====
Rescue Shell

Your system has been mounted under /mnt/sysroot.

If you would like to make the root of your system the root of the active system,
run the command:

    chroot /mnt/sysroot

Warning: The rescue shell will trigger SELinux autorelabel on the subsequent
boot. Add "enforcing=0" on the kernel command line for autorelabel to work
properly.
When finished, please exit from the shell and your system will reboot.

Please press ENTER to get a shell:
bash-5.1# chroot /mnt/sysroot
bash-5.1# dracut --regenerate-all --force
bash-5.1# exit
exit
bash-5.1# exit_
```

As you can see the remove the file system are restored and machine works normal

```
Password:
[root@client ~]# cd /boot
[root@client boot]# ll
total 487564
-rw-r--r--. 1 root root 240333 Apr 22 14:44 config-5.14.0-697.el9.x86_64
-rw-r--r--. 1 root root 240450 May 27 14:43 config-5.14.0-710.el9.x86_64
drwxr-xr-x. 3 root root 17 Jun 10 15:04 efi
drwx-----. 5 root root 97 Jun 26 19:49 grub2
-rw-----. 1 root root 268647766 Jun 10 15:13 initramfs-0-rescue-e62ba35a032f4beb951d75b1feef762.img
-rw-----. 1 root root 62983352 Jun 27 07:43 initramfs-5.14.0-697.el9.x86_64.img
-rw-----. 1 root root 62996268 Jun 27 07:44 initramfs-5.14.0-710.el9.x86_64.img
-rw-----. 1 root root 39386112 Jun 27 07:48 initramfs-5.14.0-710.el9.x86_64kdump.img
drwxr-xr-x. 3 root root 21 Jun 10 15:08 loader
lrwxrwxrwx. 1 root root 45 Jun 10 15:10 symvers-5.14.0-697.el9.x86_64.gz -> /lib/modules/5.14.0-697.el9.x86_64/symvers.gz
lrwxrwxrwx. 1 root root 45 Jun 22 11:17 symvers-5.14.0-710.el9.x86_64.gz -> /lib/modules/5.14.0-710.el9.x86_64/symvers.gz
-rw-----. 1 root root 9561583 Apr 22 14:44 System.map-5.14.0-697.el9.x86_64
-rw-----. 1 root root 9566527 May 27 14:43 System.map-5.14.0-710.el9.x86_64
-rwxr-xr-x. 1 root root 15206440 Jun 10 15:11 vmlinuz-0-rescue-e62ba35a032f4beb951d75b1feef762
-rwxr-xr-x. 1 root root 15206440 Apr 22 14:44 vmlinuz-5.14.0-697.el9.x86_64
-rwxr-xr-x. 1 root root 15214632 May 27 14:43 vmlinuz-5.14.0-710.el9.x86_64
[root@client boot]#
```

Kernel – initramfs file remove

After removing initramfs file I would get a error like this

Reboot or shutdown then power on

Now power on the machine with restart guest and press esc while screen splash

Click the cd rom drive

Click or select the troubleshooting to the grub error

Then select the resure a centos stream system

It will take us to prombt where start 1 to continue

Then press enter to get shell

Type the command

```
#Chroot /mnt/sysroot
```

```
#Dracut --regenerate-all --force
```

```
#exit
```

```
#exit
```

As you can see the removed file system are restored and machine works normal

The initramfs file is required by the Linux kernel during the boot process. It contains the drivers and files needed to mount the real root file system. If the initramfs file is deleted, the system cannot boot. To fix this, I boot the server using the CentOS installation DVD or ISO, select **Troubleshooting**, and then **Rescue a CentOS Stream system**. I choose option **1** to mount the installed operating system and run `chroot /mnt/sysroot` to work inside the installed system. Then I regenerate the initramfs image by running `dracut --regenerate-all --force`. Finally, I exit the rescue environment, remove the installation media, reboot the system, and verify that the initramfs file is present in `/boot`. The system then boots normally.